



User Manual

cAMP Hunter™ GIP RA Bioassay Kit

For the Measurement of Ligand-Mediated cAMP Accumulation

For Bioassay Kits with control

95-0146Y2-00204: 2-Plate Kit

95-0146Y2-00205: 10-Plate Kit

For bioassay Kits without control

95-0146Y2-00206: 10-Plate Kit



Document Number 70-449 Revision 1

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1. Overview

The cAMP Hunter GIP RA Bioassay Kit provides a robust and highly sensitive functional, cell-based assay to monitor 3'-5'-cyclic adenosine monophosphate (cAMP) production in cells as a result of ligand-mediated Gastric Inhibitory Polypeptide (GIP) Receptor Activation. The bioassay kit contains all the materials needed for a complete assay, including cells, detection reagents, cell plating reagent, positive control agonist, and assay plates. Note that a kit is also available with all reagents listed above except the positive control ligand. The ready-to-use cryopreserved cells included in the kit have been manufactured to ensure assay reproducibility and faster implementation from characterization to lot release. This bioassay has been optimized for a 96-well format but can be adapted to a 384-well format as well.

2. Assay Principle

Ligand-mediated G-Protein Coupled Receptor (GPCR) stimulation leads to the activation of G-proteins, which in turn triggers downstream signaling pathways by recruiting, activating or inhibiting cellular enzymes. One such enzyme is adenylate cyclase, which catalyzes the conversion of adenosine triphosphate (ATP) to cAMP. Adenylate cyclase is either stimulated or inhibited by the G-protein subunits G_{α_s} and G_{α_i} , respectively. The cAMP Hunter GIP RA Bioassay monitors activation of the naturally G_{α_s} -coupled receptor, GIPR, by measuring the cellular cAMP levels using a homogeneous, gain-of-signal competitive immunoassay based on Enzyme Fragment Complementation (EFC) technology, which is explained in the [cAMP Detection Kit Principle](#) section.

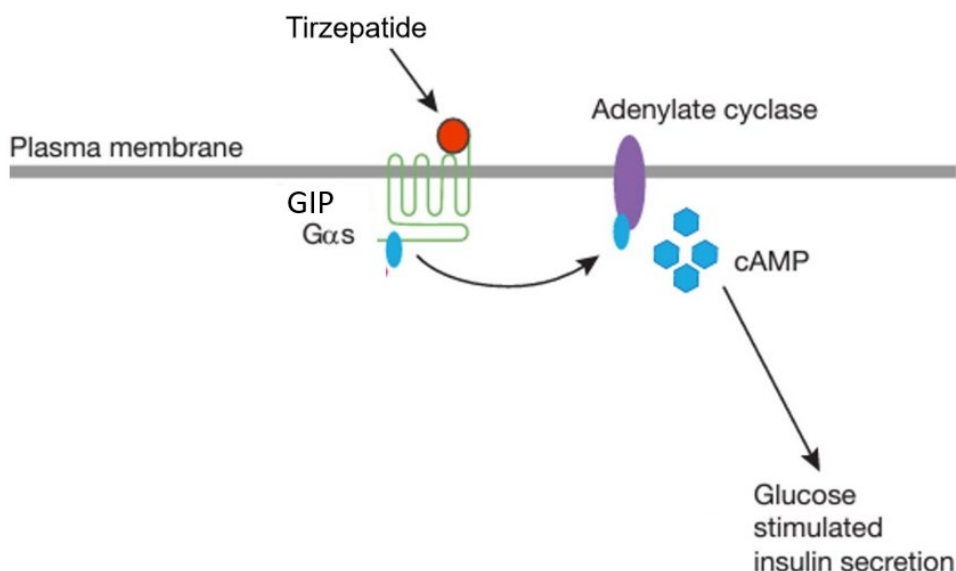


Figure 1. Assay Principle

The cAMP Hunter Bioassay has been developed to interrogate the GPCR cAMP Pathway. Ligand-mediated activation of a GPCR either stimulates or inhibits adenylate cyclase to modulate cellular cAMP levels. In the case of GIPR, its activation by GIP stimulates adenylate cyclase, which in turn enables the production of cAMP.

3. cAMP Detection Kit Principle

The EFC technology uses a β -galactosidase (β -gal) enzyme split into two fragments, the Enzyme Donor (ED) and Enzyme Acceptor (EA). Independently these fragments have no β -gal enzymatic activity; however, in solution they rapidly complement to form an active β -gal enzyme.

In this assay, cAMP from cell lysates and ED-labeled cAMP (ED-cAMP) compete for anti-cAMP antibody (Ab). Antibody-bound ED-cAMP does not complement with EA, but unbound ED-cAMP is free to complement with EA to form an active enzyme, which subsequently produces a chemiluminescent signal. The signal from the assay is directly proportional to the amount of cellular cAMP in the well, i.e., the higher the GPCR activation, the greater the cAMP levels inside the cells, and larger the signal in the assay.

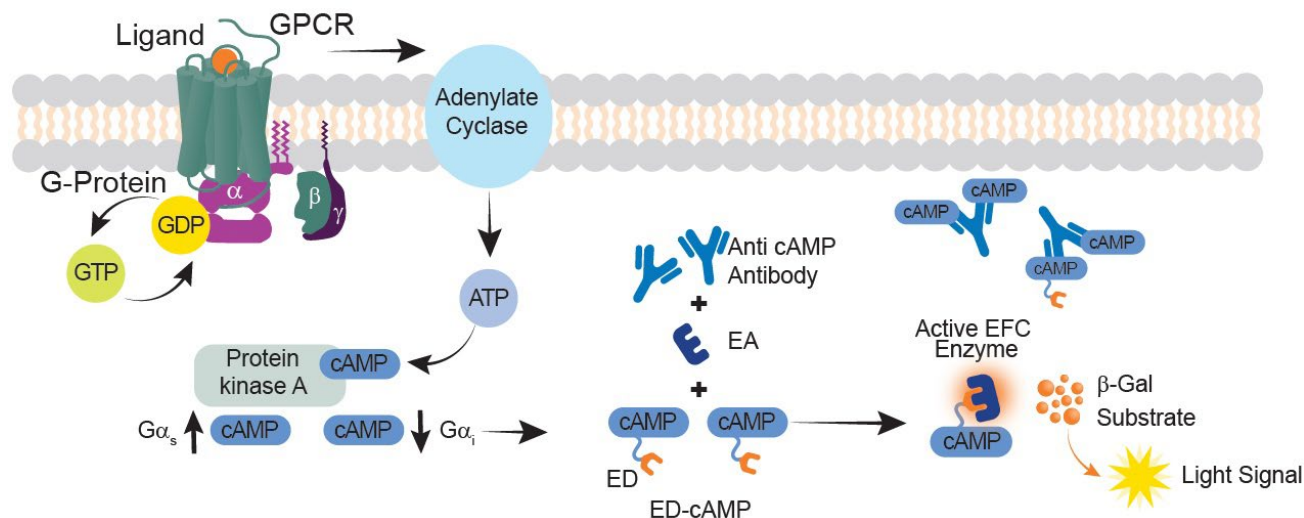


Figure 2. cAMP Detection Kit Principle

When cellular cAMP levels are low, ED-labeled cAMP successfully binds with the anti-cAMP-antibody, as there aren't enough cellular cAMP molecules to compete against it. The bound ED-labeled cAMP will not be available to undergo complementation. In contrast, when cellular cAMP levels are high, the cAMP molecules bind to the anti-cAMP antibody instead, leaving ED-labeled cAMP relatively free. Upon the addition of the detection reagent containing EA, the ED and EA fragments undergo complementation, which successfully forms an active β -galactosidase enzyme that hydrolyzes the substrate and generates a chemiluminescent signal proportional to the level of cAMP in the cell.

4. Materials Provided

List of Components	95-0146Y2-00204 (2-Plate Kit)	95-0146Y2-00205 (10-Plate Kit)	95-0146Y2-00206 (10 plate Kit without control)
cAMP Hunter CHO-K1 GIPR Bioassay Cells (2.4×10^6 cells in 0.2 mL per vial)	2 Vials	10 Vials	10 Vials
AssayComplete™ Cell Plating 2 Reagent (CP2) (100 mL per bottle)	1 x 100 mL	2 x 100 mL	2 x 100 mL
AssayComplete Cell Assay Buffer (50 mL per bottle)	1 x 50 mL	2 x 50 mL	2 x 50 mL
Control Agonist (GIP), 500 µg	1 x 500 µg	1 x 500 µg	N/A*
cAMP Detection Kit for Bioassays			
cAMP Standard (250 µM) (Bottle)	1 x 0.2 mL	1 x 1 mL	1 x 1 mL
cAMP Antibody Reagent (Bottle)	1 x 5 mL	1 x 25 mL	1 x 25 mL
cAMP Lysis Buffer (Bottle)	1 x 7.6 mL	1 x 38 mL	1 x 38 mL
Substrate Reagent 1 (Bottle)	1 x 2 mL	1 x 10 mL	1 x 10 mL
Substrate Reagent 2 (Bottle)	1 x 0.4 mL	1 x 2 mL	1 x 2 mL
cAMP Solution D (Bottle)	1 x 10 mL	1 x 50 mL	1 x 50 mL
cAMP Solution A (Bottle)	1 x 16 mL	1 x 80 mL	1 x 80 mL
96-well White, Flat-bottom Poly-D-Lysine-coated, Sterile Plates with Lid	2 Plates	10 Plates	10 Plates

***Note:** For 95-0146Y2-00206, control agonist is not provided in the kit and would need to be obtained separately if needed. Refer to additional materials required table for ordering information for the control agonist.

Storage Conditions

cAMP Hunter CHO-K1 GIPR Bioassay Cells

Cells are shipped on dry ice and should arrive in a frozen state. To ensure maximum cell viability, store the vials of bioassay cells in the vapor phase of liquid nitrogen as soon as possible upon receipt. Please contact technical support immediately, if the cells received were already thawed.

- Short-term (24 hours or less): Store vials at -80°C immediately upon arrival. (DO NOT store at -80°C for longer than 24 hours).
- Long-term (greater than 24 hours): Vials should ONLY be stored in the vapor phase of liquid nitrogen.



Safety Warning: A face shield, gloves and lab coat should be worn at all times when handling frozen vials. Use tongs to remove cryovials from liquid nitrogen storage and place the vials immediately on dry ice in a covered container. Wait at least 1 minute for any liquid nitrogen that may be present inside the vial to evaporate. Do not touch the bottom of the vials at any time to avoid inadvertent thawing of the cells.

AssayComplete™ Cell Plating 2 Reagent (CP2)

Upon receipt, store at -20°C. Once thawed, the Cell Plating 2 Reagent can be stored at 4°C for up to 4 weeks. For longer storage (up to the expiration date listed on the kit's Certificate of Analysis), the reagent should be aliquoted and stored at -20°C. Once aliquoted, do not freeze-thaw more than two times. Make aliquots suitable for testing one assay plate, for example: 20 mL of reagent per aliquot can be dispensed and stored at -20°C. Ensure that the reagent is equilibrated to room temperature before use in the assay for best performance.

AssayComplete Cell Assay Buffer

Upon receipt, store at -20°C. Once thawed, the Cell Assay Buffer can be stored at 4°C for up to 4 weeks. For longer storage (up to the expiration date listed on the kit's Certificate of Analysis), the reagent should be aliquoted and stored at -20°C. Once aliquoted do not freeze-thaw more than twice. Make aliquots suitable for testing one assay plate, for example: 10 mL of reagent per aliquot should be dispensed and frozen down. This amount may vary depending on your stock sample concentrations and should be adjusted accordingly. Ensure that the reagent is equilibrated to room temperature before use in the assay for best performance.

Control Agonist (GIP)

Upon receipt, store at -20°C until ready to use (up to the expiration date listed on the kit's Certificate of Analysis). Centrifuge the vial prior to opening, to maximize recovery, and reconstitute as recommended in the ligand datasheet with 0.200 mL of tissue culture grade water (as recommended by vendor) to make a stock concentration of 500 µM. Store in small, single-use aliquots at -20°C. The reconstituted ligand is stable for 12 months at -20 to -80°C, or 1 week at 2-8°C.

cAMP Detection Kit for Bioassays

Upon receipt, store the kit at -20°C. Once thawed, the kit components can be stored at 4°C for up to 4 weeks. For longer storage (up to the expiration date listed on the kit's Certificate of Analysis), the reagents should be aliquoted and stored at -20°C in opaque containers until needed. Once aliquoted do not freeze-thaw more than twice. Make aliquots suitable for testing one assay plate. This amount may vary depending on your assay set-up and should be adjusted accordingly. Ensure that the reagents are equilibrated to room temperature before use in the assay for best performance.

96-well White, Flat-bottom Poly-D-Lysine-coated, Sterile Plates with Lid

Upon receipt, store at room temperature.

5. Additional Materials Required

The below equipment and additional materials are required to perform these assays. Equivalent reagents may be used. All plastic materials should be stored at temperatures specified by suppliers.

Material	Ordering Information
*Control Agonist (GIP)	DiscoverX, Cat No. 92-1078, or similar
96-Well Green, V-Bottom, Untreated, Non-Sterile Dilution Plates	DiscoverX, Cat No. 92-0011, or similar
Disposable polystyrene reagent reservoirs (25mL), sterile	Thermo Fisher Scientific (Cat.#. 8094), or similar
15 mL LightSafe polypropylene tubes, sterile	Millipore Sigma (Cat # Z688320), or similar
Multimode or luminescence plate reader	Refer to the Instrument Compatibility Chart at discoverx.com/instrument-compatibility
50 mL and 15 mL Polypropylene tubes, sterile	
1.5 mL polypropylene microcentrifuge tubes, sterile	
Tissue culture disposable pipettes (1 mL - 25 mL), sterile	
Disposable pipet tips for P20, P100, P1000 pipettes	
Sterile biosafety cabinet	
Automated cell counter or hemocytometer	
Humidified tissue culture incubator	
Single and multichannel pipettes and pipette tips (10 µL-1000 µL)	

Note: *Positive control used for QC testing of the bioassay cells in this kit as reflected in Certificate of Analysis; only needed if the kit purchased was the variant without control included

6. Unpacking Cell Cryovials

Cryovials are shipped on dry ice and contain cells in 0.2 mL of AssayComplete™ Freezing Reagent. Upon receipt, the vials should be transferred to the vapor phase of liquid nitrogen for storage beyond 24 hours. The following procedures are for the safe storage and removal of cryovials from liquid nitrogen storage.



Appropriate personal protective equipment (PPE), including a face shield, gloves, and a lab coat should be worn during these procedures.

1. cAMP Hunter CHO-K1 GIPR Bioassay Cells must arrive in a frozen state on dry ice.



Contact technical support immediately, if cells received were already thawed.

2. Frozen cells must be transferred to either liquid nitrogen storage or a -80°C freezer immediately upon arrival. If cells will be thawed and used within 24 hours, they can be stored temporarily at -80°C. For long-term storage, place vials in the vapor phase of liquid nitrogen storage.



Cryovials should be stored in the vapor phase of liquid nitrogen ONLY. These vials are not rated for storage in the liquid phase.

3. Using tongs, remove the cryovials from the liquid nitrogen storage, and place them immediately in dry ice, in a covered container. Wait for at least one minute for any liquid nitrogen inside the vial to evaporate.
4. Proceed with the thawing protocol in the detailed assay protocol section in section 8.1.

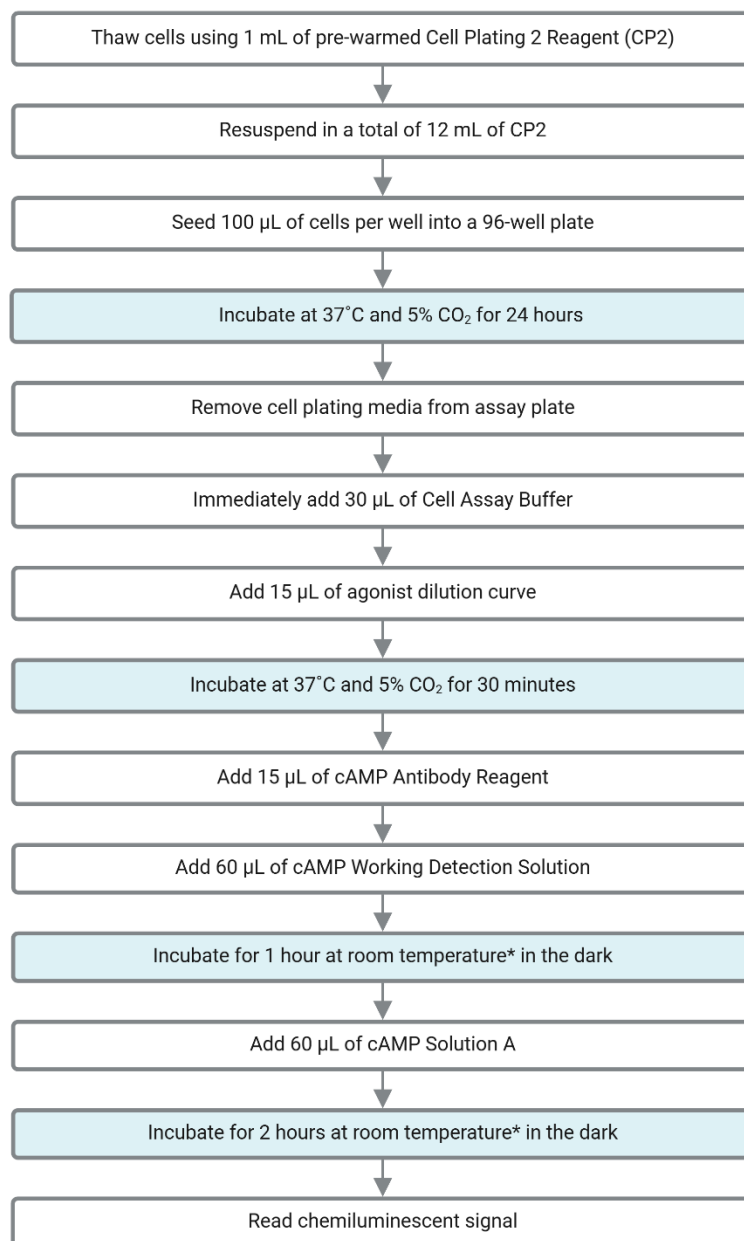
7. Protocol Schematic

Quick-start Procedure: In a 96-well plate, perform the following steps.

Tip: Use this sheet to note your assay specific conditions. Post it on your bench to use as a quick reference guide.

Assay Name: _____ Date: _____

Product Details: _____



*Room temperature refers to a range of 23-25°C

8. Detailed Protocol

This assay is performed under aseptic conditions. Preparation of cells, Reference Standards/Test Samples are performed in a Biological Safety Cabinet following good aseptic technique. All appropriate materials are either certified sterile or prepared aseptically.

Prepared volumes may be scaled up or down if required. This user manual provides a protocol for determining potency in a 96-well format.

8.1 Bioassay Cell Preparation

(Day 1)

The following protocol is for thawing and plating cryopreserved CHO-K1 GIPR bioassay cells from cryovials.

1. Before thawing the cells, ensure that all the materials required are set up in the biosafety cabinet. These include:
 - a. One sterile 25 mL reagent reservoir
 - b. One sterile 15 mL conical tube
 - c. A pipette (P1000) set to dispense 1 mL
 - d. A multichannel pipette and tips set to dispense 100 µL
 - e. An aliquot of AssayComplete™ Cell Plating 2 Reagent (CP2), pre-warmed in a 37°C water bath for 15 minutes and equilibrated to room temperature.
 - f. A 96-well White, Flat-bottom Poly-D-Lysine-coated, Sterile Plates with Lid (provided with the kit)
2. Dispense 12 mL of CP2 into the 15 mL conical tube.
3. Remove the cryovial from the liquid nitrogen vapor and immediately place it in dry ice.
4. Remove the cryovial from dry ice and ensure cap is tightened. Immediately thaw vial in 37°C water bath for 30 (+/- 5) seconds.
DO NOT LEAVE THE VIAL IN WATER BATH.
5. Visually inspect bottom of vial after 20 seconds. If pellet is thawed, remove vial from water bath, wipe down the outside surface of the cryovial quickly with 70% ethanol, and immediately bring it into the biosafety cabinet. If ice is still visible, return vial to water bath for additional 10-15 seconds.
6. Add 1 mL of pre-warmed CP2 from the 15 mL conical tube into the cryovial. The reagent should be added slowly along the wall of the cryovial tube. Slowly pipet the cell suspension up and down, several times, to uniformly resuspend the cells.
7. Transfer the cell suspension to the conical tube containing the remaining 11 mL of CP2. Remove any remaining liquid from the cryovial to ensure maximum recovery of all the cells. Mix by gently pipetting cell suspension up and down 2-3 times with a 10 mL serological pipet.
8. Remove an aliquot (e.g. 20 µL) of cell suspension and count cells using a hemacytometer or automated counter. Adjust cell density to 0.2×10^6 cells/mL with pre-warmed CP2 as needed.
9. Mix cell suspension in conical tube by gentle inversion 2-3 times to ensure that the cells are properly

resuspended in the reagent, without creating any froth in the suspension. Immediately transfer the suspension into the sterile 25 mL reagent reservoir.

10. Using a manual 12-channel multichannel pipette, transfer 100 µL of the cell suspension to each well of the 96-well assay plate (flat-bottom Poly-D-Lysine-coated), one row at a time, using reverse pipetting. Mix cells in reagent reservoir by pipetting up/down 2-3 times before aspirating and dispensing 100 µL cells into each subsequent row in the assay plate.
11. Replace the lid and leave the plate at room temperature for 15 minutes, to allow the cells to settle uniformly in the well and minimize potential edge effects.
12. Incubate the assay plate in humidified tissue culture incubator at 37°C and 5% CO₂ for 24 ± 2 hours before proceeding with the assay.

8.2 GIP Control Ligand and Test Sample Preparation

(Day 2)

The following protocol is the recommended dilutions for preparing the GIP control ligand and test sample working stocks and serial dilutions.

1. Before beginning, ensure all required reagents are assembled in biosafety cabinet:
 - a. Micropipettes (P20, P100, P1000).
 - b. A multichannel pipette and tips set to dispense 20 µL.
 - c. An aliquot of AssayComplete™ Cell Assay Buffer, pre-warmed in a 37°C water bath for 15 minutes, then equilibrated to room temperature.
 - d. A 96-Well Green, V-Bottom, Untreated, Non-Sterile Dilution Plate, labeled 'Master dilution plate' (MDP)
 - e. Sterile microfuge tubes, for preparation of intermediate, working, or nominal testing concentrations of GIP control ligand or test samples.
2. On day of assay, prepare GIP ligand intermediate and working stock solutions in Cell Assay Buffer (CAB) as shown in examples in Table 1 below.
 - a. GIP control ligand (see [Additional Materials Required](#) table) is supplied as a lyophilized powder and is dissolved in 0.200 mL of tissue culture grade water (as recommended by vendor) to prepare a 500 µM stock (or 2.5 mg/mL) concentration. To avoid condensation, equilibrate the vial to ambient temperature before opening.

Note: Reconstituted GIP ligand stock from Step 2a may be stored in small aliquots at -20°C for up to 12 months (as specified in the product specific datasheet). Please avoid multiple freeze/thaw cycles.

- b. It is recommended to use a minimum aliquot size of 10 µL for preparation of all sample / stock dilutions.

Table 1. Example Preparation of GIP Intermediate Dilutions & Working Stocks

Intermediate and Working Stock Solutions	Final Concentration, $\mu\text{g/mL}$	Volume of GIP (Stock concentration)	Volume Dilution Buffer (CAB), μL
GIP ligand, Intermediate Stock 1	125.0	10 μL of 2500 $\mu\text{g/mL}$	190
GIP ligand, Intermediate Stock 2	12.5	20 μL of 125.0 $\mu\text{g/mL}$	180
GIP ligand, Intermediate Stock 3	1.25	20 μL of 12.5 $\mu\text{g/mL}$	180
Working Stock of GIP	0.600	144 μL of 1.25 $\mu\text{g/mL}$	156

Note: Use a vortex to mix (10-20 sec) for each dilution before preparing next serial dilution.

3. On the day of assay, prepare serial dilutions of the GIP ligand in row A of the 96-well master dilution plate (MDP), at 3X the final concentrations of each dilution, in CAB (i.e. the Ligand Diluent) as per Table 2 below. Sufficient volumes to run triplicate wells per dose in the assay plate will be the result.

- Add appropriate volume of ligand diluent (CAB) to row A of the MDP, (as indicated in column 6) of Table 2.
- Transfer indicated volume of Working Stock of GIP to the MDP Row A Well 2; the volume that should be transferred is indicated in column 5 of Table 2.
- Prepare the dilution series by transferring the volume of GIP ligand (indicated in column 5) from the source well (indicated in column 4) to the destination well (indicated in column 1). Pipet up and down several times to mix in destination wells. Replace pipet tips between each serial dilution. No sample is added to well 1 (vehicle only), as this serves as the negative control.

Table 2. GIP Control Ligand concentration in Master Dilution Plate prepared as a 3X stock: Example of Preparation of GIP Ligand Serial Dilutions

Column 1	Column 2	Column 3	Column 4	Column 5	Column 6	Column 7
Well location on Master Dilution Plate (MDP)	Concentration (3X) of GIP on MDP Row A, ng/mL	Dilution Factor	Dilution (3X) GIP Source Well	Volume (3X) of GIP added, μ L	Volume of Ligand Diluent, μ L	Final Concentration (1X) of GIP in Assay Plate, ng/mL
Row A, Well 1	0	--	--	--	180	--
Row A, Well 2	600		Working Stock (600 ng/mL; see Table 1)	300	-	200
Row A, Well 3	200	3	Row A, Well 2	100	200	66.667
Row A, Well 4	66.667	3	Row A, Well 3	100	200	22.222
Row A, Well 5	22.222	3	Row A, Well 4	100	200	7.407
Row A Well 6	11.111	2	Row A, Well 5	150	150	3.704
Row A, Well 7	5.556	2	Row A, Well 6	150	150	1.852
Row A, Well 8	1.852	3	Row A, Well 7	100	200	0.617
Row A, Well 9	0.617	3	Row A, Well 8	100	200	0.206
Row A, Well 10	0.206	3	Row A, Well 9	100	200	0.069
Row A, Well 11	0.069	3	Row A, Well 10	100	200	0.023
Row A, Well 12	0.023	3	Row A, Well 11	100	200	0.008

4. On the day of assay, prepare serial dilutions of the test sample in row B of the 96-well master dilution plate (MDP), at 3X the final concentrations of each dilution, in CAB, as illustrated in Table 2 for GIP control ligand.

Note: Depending on sample concentration, intermediate dilutions of test sample(s) may need to be prepared (as demonstrated in Table 1 for GIP control ligand) to generate appropriate concentration of working stock needed to prepare top dose in dilution series.

5. Assay plate preparation:
- Remove assay plates with cells from incubator.
 - With a flicking motion, remove liquid from the assay wells in the plate. Then, place the assay plate upside down on a lab-grade low lint tissue/paper towel and spin in a plate centrifuge set at 1000 rpm. Stop once it reaches 1000rpm. Alternately, completely remove the Cell Plating Reagent from each assay plate by carefully aspirating the wells.
 - Immediately add 30 μ L of Cell Assay Buffer to all wells of the plate.

6. Transfer 15 μ L of the 3X sample dilutions from the MDP to the appropriate wells containing cells in the assay plate. Refer to **Figure 3. Representative Assay Plate Map**:
 - a. Row A in MDP: transfer to Rows B, D and F in the assay plate (GIP control agonist)
 - b. Row B in MDP: transfer to Rows C, E and G in the assay plate (Test Sample)
7. Incubate the assay plate in a humidified incubator at 37°C and 5% CO₂ for 30 minutes.

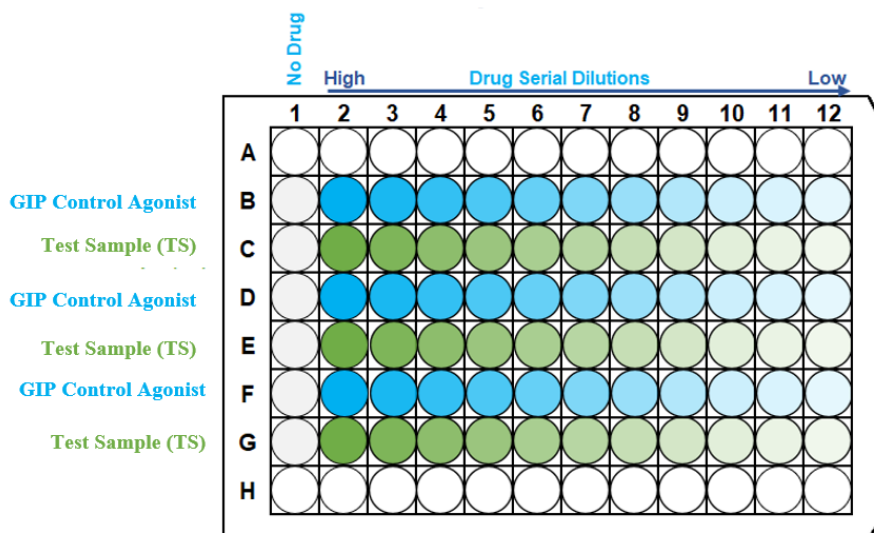


Figure 3. Representative Assay Plate Map

This plate map shows two interdigitated 11-point dose curves, with three replicates per dose point, for a test sample (TS) and GIP control agonist) prepared using the same dilution scheme. Column 2 contains the highest dose of each sample, while column 12 contains the lowest dose. Column 1 contains no drug (CAB only).

8.3 Addition of Detection Reagent

Signal Detection

1. During the 30-minute agonist incubation period, prepare cAMP working detection solution in a separate 15 mL polypropylene tube by mixing 19-parts of cAMP Lysis Buffer, 5-parts of Substrate Reagent 1, 1-part Substrate reagent 2, and 25-parts of cAMP Solution D. Refer to Table 5 below for the volume of each component required for one 96-well plate. Adjust volumes according to your assay set-up, keeping the ratios consistent. Store in the dark before use.
2. Following agonist incubation, add 15 μ L of the cAMP Antibody Reagent to all wells of the assay plate.

Table 5. Working Detection Solution Preparation

Components	Volume Ratio	Volume per 96-well Plate (mL)
cAMP Lysis Buffer	19	3.8
Substrate Reagent 1	5	1.0
Substrate Reagent 2	1	0.2
cAMP Solution D	25	5.0
Total Volume		10

3. Add 60 μ L of the prepared cAMP working detection solution to all wells of the assay plate. **Do not pipette up and down or vortex the plate to mix.**
4. Incubate the assay plate for 1 hour at room temperature in the dark.

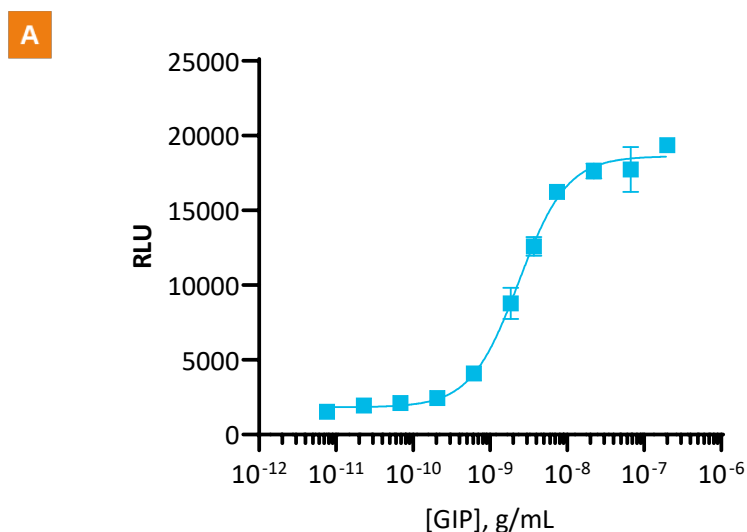


Room temperature refers to a range of 23-25°C.

5. Add 60 μ L of cAMP Solution A to all wells of the assay plate. **Do not pipette up and down or vortex the plate to mix.**
6. Incubate the assay plate for 2 hours at room temperature in the dark.
7. Read the sample on a Perkin Elmer Envision with a 0.2 sec/well integration time, or on a similar bench top instrument. Refer to the Instrument Compatibility Chart at discoverx.com/instrument-compatibility/
8. Data analysis can be performed using any statistical analysis software such as GraphPad Prism, SoftMax Pro, Gen5, or Microsoft Excel etc.

9. Typical Results

The following graph in Figure 4 is an example of a typical dose-response curve for the cAMP Hunter GIP RA Bioassay generated using the protocol outlined in this manual. The data shows a potent, dose-dependent increase in cAMP production when the bioassay cells were treated with GIP ligand. The plate was read on the EnVision® Multimode Plate Reader, with a 0.2 sec/well integration time, and data analysis was conducted using GraphPad Prism.



B

S/B	EC ₅₀ , ng/mL
12.7	2.358

Figure 4. Typical Results. Representative **A**, dose-response curve and **B**, EC₅₀ and assay window (S/B) for GIP mediated GIPR activation, as measured in this bioassay.

The following graph in Figure 5 is an example of a typical dose-response curve for Tirzepatide (over a working range of 50 ng/mL to 9.3 pg/mL) in the GIP RA Bioassay generated using the protocol outlined in this manual. The plate was read on the EnVision® Multimode Plate Reader, with a 0.2 sec/well integration time, and data analysis was conducted using GraphPad Prism.

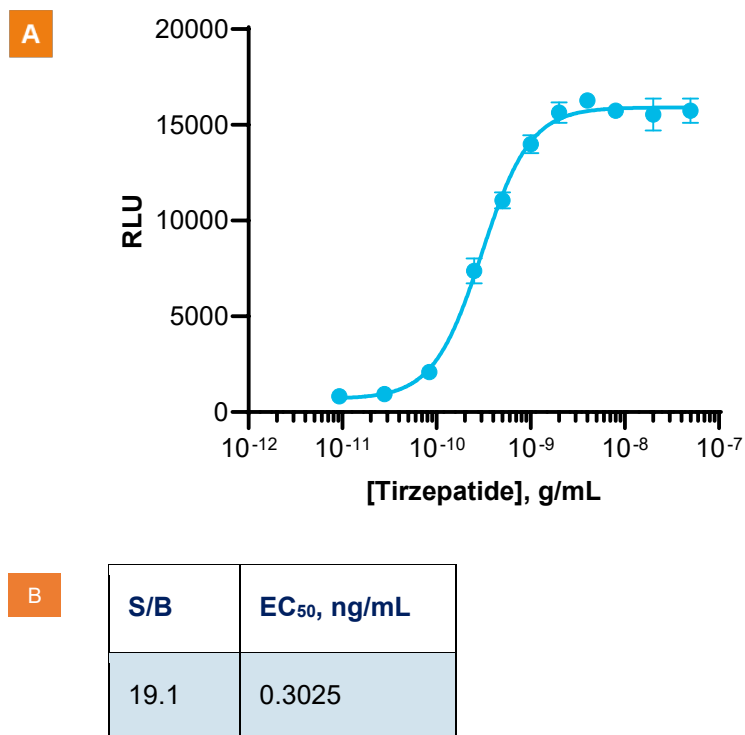


Figure 5. Typical Results. Representative A, dose-response curve and B, EC₅₀ and assay window (S/B) for Tirzepatide-mediated GIPR activation, as measured in this bioassay.

10. Troubleshooting Guide

Problem	Potential Cause	Proposed Solution
No response	Incorrect thawing procedure	Refer to the thawing instructions in the Bioassay Cell Preparation section of this user manual.
	Incorrect ligand used or improper ligand incubation time	Refer to the Certificate of Analysis for recommended ligand and assay conditions.
	Incorrect preparation of the ligand (agonist or antagonist)	Refer to the vendor-specific datasheet to ensure proper handling, dilution, and storage of the ligand.
	Sub-optimal time course for induction	Optimize time course of induction with agonist and antagonist.
Decreased or no response	Cells are not adherent and exhibit incorrect morphology	Confirm adherence of cells using a microscope.
	Problem with microplate reader	The microplate reader should be in luminescence mode. Read at 0.1-1 seconds/well.
Low or no signal	Incorrect preparation of detection reagents	Detection reagents are sensitive to light and should ideally be prepared just prior to use.
	Problem with microplate reader	The microplate reader should be in luminescence mode. Read at 0.1-1 second/well.
Experimental S/B does not match the value noted in the Certificate of Analysis provided	Incorrect incubation temperature	Confirm assay conditions.
		Check and repeat the assay at the correct incubation temperature, as indicated on the assay datasheet.
	Incorrect preparation of ligand (agonist or antagonist)	Some ligands are difficult to handle. Confirm the final concentration of ligands.
EC ₅₀ is right-shifted	Improper ligand handling or storage	Ensure that the ligands are stored and incubated at the proper temperature.
	Difference in agonist binding affinity	Confirm that the ligand used is comparable to the ligand in the Certificate of Analysis.
	Problems with dynamic range or dilutions	Changing tips during dilution can help in avoiding carryover.
		Include a cAMP standard curve in the assay to ensure that the ligand dose curve is in the dynamic range of the cAMP detection kit. Moving out of the dynamic range may shift the EC ₅₀ of the ligands.
High variability between replicates	Instrument calibration	Ensure that the dispensing equipment is correctly calibrated, and proper pipetting technique is used.

For questions on using this product, please contact Technical Support at discoverx.com/support/

Document Revision History

Revision Number	Date Released	Revision Details
0	April 2024	New Document
1	March 2025	Corrected cell reconstitution volume, cell number / vial and added cell counting step

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